

● Symposium V—Clinical Charged Particles

EVALUATION OF THE CLINICAL APPLICABILITY OF PROTON BEAMS IN DEFINITIVE FRACTIONATED RADIATION THERAPY

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We report on the treatment of 317 patients treated either wholly or in part with proton beams at the Harvard Cyclotron Laboratory. These include: 130 patients treated for definitive radiation therapy of choroidal melanoma; 17 patients treated for tumors of the base of skull, cervical spine and cranium, which abut structures of the central nervous system (CNS); 23 patients treated for sarcomas of soft tissue and bone; 65 patients treated for carcinoma of the prostate; 14 patients treated for carcinoma of the rectum and anus; and 23 patients treated for squamous carcinoma of the oral cavity and oro-pharynx. Data on causes of failure and morbidity of treatment are presented. Overall the results are judged to be extremely encouraging. In particular, the treatment of the choroidal melanomas and sarcomas abutting CNS structures have clear clinical value, and the treatment of prostatic tumors and tumors of the head and neck are thought to be promising.

Proton, Particle, Melanoma, Chordoma.

INTRODUCTION

The experience of our Massachusetts General Hospital—Harvard Cyclotron Laboratory (MGH-HCL) group is based on 317 cancer patients who have completed definitive radiation therapy that employed proton beam techniques either for the boost dose or for the entirety of the treatment. This paper is a sequel to our report to the Part II Conference in 1976, which was based upon 23 patients.³ Higher radiation doses have been administered with the proton beam than we would have used with conventional photon and/or electron beam techniques. These higher doses have been well tolerated; and marginal misses have been quite uncommon. The immobilization and position confirmation techniques permit precision of repositioning of patients to sub-millimeter levels for treatment of choroidal melanoma and ~1 mm precision for selected lesions such as those abutting on the CNS. Results in terms of local control and freedom from complication conform to the predicted benefits from use of superior dose distributions. Finally, our results strongly encourage us to evaluate more energetic, i.e., more deeply penetrating, proton beams for the treatment of lesions

more deeply sited than those which are feasible with the 160 MV proton beam available to us.

RELATIVE BIOLOGICAL EFFECTIVENESS

We have measured the relative biological effectiveness (RBE) of protons using a variety of *in vivo* endpoints and have been fortunate to have several scientists from other laboratories come to the Harvard Cyclotron and perform RBE determinations using *in vitro* and *in vivo* assay methods. The results of these studies have recently been reviewed.⁴ The RBE values have ranged from slightly less than 1.0 to 1.42 in the various assays. The modal value for low dose levels is judged to lie in the range of about 1.15 to 1.20. For our clinical proton radiation therapy we have used an RBE value of 1.10; this is a conservative value (to avoid under-dose) and our clinical observations are more consistent with a value of 1.10 than 1.20.

RESULTS OF PROTON BEAM RADIATION THERAPY

Table 1 presents a list of the number of patients who have been treated for tumors at various anatomic sites. We now discuss our experience with these treatments.

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